SOFC PLANT · WALKTHROUGH PAGE 6 CHECKED AGAINST THE SENSORS

Mostly yes. The gas-line heaters, the burner and the air pre-heaters need fixing.

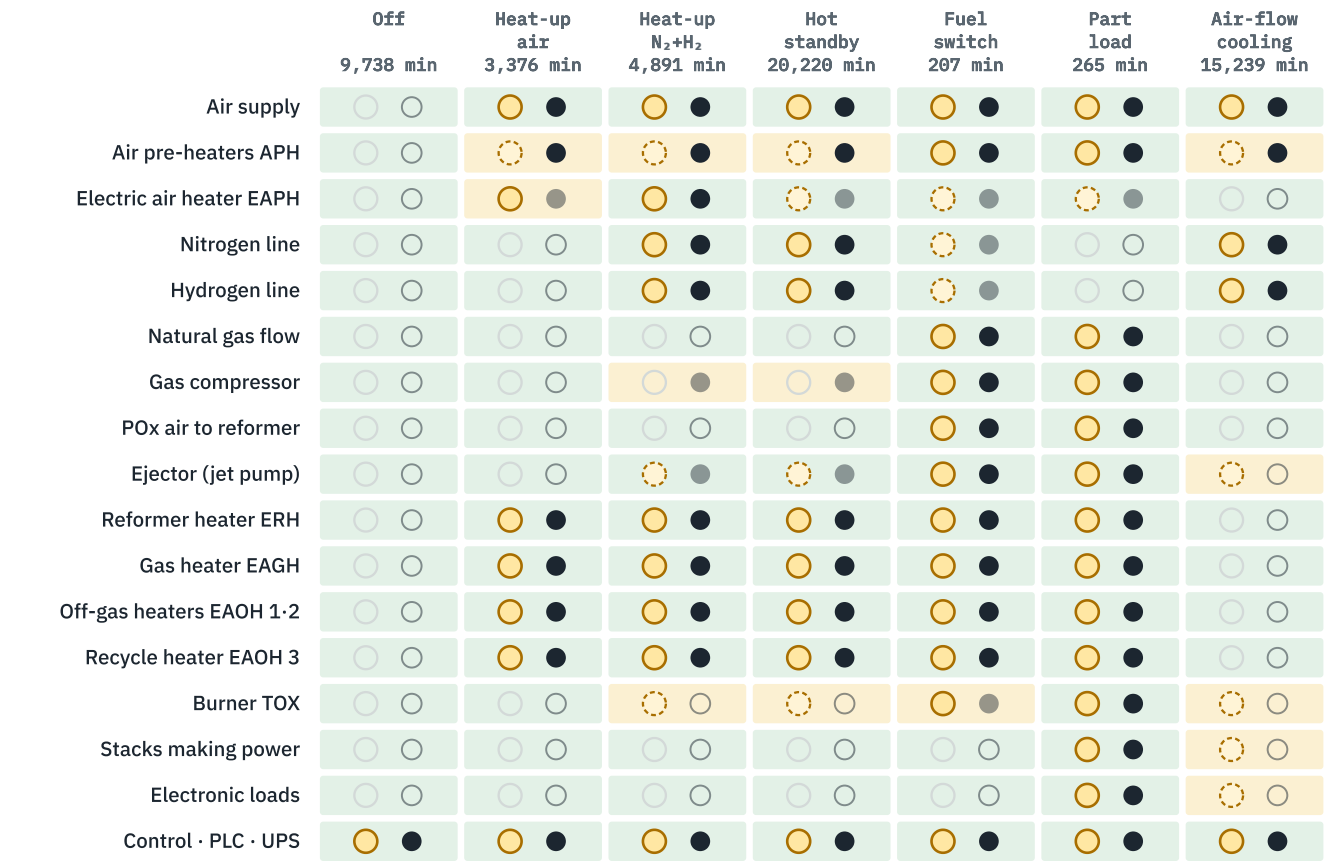
Page 6 of the plant walkthrough says which devices run in each plant stage. We checked every one of those claims against the logged sensor readings for the same minutes.

> All the gas and air supplies, the stacks and the loads behave exactly as the page says. Three groups do not: the **small gas-line heaters** are already running in the first start-up step, the **air pre-heaters** heat the air in every stage that has air flowing, and the **burner** shows no sign of burning while only forming gas is flowing.

105	14	0
of 119 cells agree	one step apart (on vs partly)	clearly wrong (on vs off)

Every device, every stage

Show only the differences



17 devices × 8 stages. Hover or tap a cell to see the share of minutes the device was running. “Partly” in the data means it ran in 10–60 % of the minutes, or ran but at low power.

What this means for page 6

- **Keep:** air, nitrogen, hydrogen, natural gas, POx air, stacks, loads, and the control system (the compressor only needs a small tweak).
- **Change:** the four small gas-line heaters, the air pre-heaters, the burner, and the air-flow cooling column. Page 2 shows each one with its numbers.
- **New stage names:** the old shut-down is now **air-flow cooling**; once the air is off the plant counts as **plant off**, and “Making power” is renamed **Part load**. Page 3 shows why, and how long the fuel switch-over lasts before power starts.
- **Fix the footnote:** page 6 says the gas-line heaters were “inferred, not measured”. They are measured: the log has their electric currents (IIR 3.1.1, 5.1.1, 5.1.2, 5.1.3, 6.1.1).
- **Safety trips:** kept out of the grid. They follow the safety design in the component sensor map: **emergency purge** and **emergency stop**. The logged trips match it (page 2, point 6).
- **Applied:** all these corrections are now in the plant walkthrough. The grid below still compares with the original page 6, so you can see what changed. Exception: the four **gas-line heater** rows (ERH, EAGH, EAOH 1·2, EAOH 3) already show the corrected page, on

from heat-up air through part load. A heater that is running counts as **on even when it runs below full power** (partly), for example ERH at about 3.2 A before natural gas (about 8 A for EAOH 3, running 72 % of heat-up-air minutes).

- **Forming-gas rule:** nitrogen and hydrogen are on whenever the stack is above 210 °C and no natural gas flows (an emergency stop overrides it). The walkthrough now uses this one rule instead of a stage-by-stage list, so both lines show on in air-flow cooling, meaning "while the stack is above 210 °C". It matches 96 % of the logged minutes. The misses are heat-ups that stayed on air a little past 210 °C (forming gas started at 190–275 °C), a few cool-downs that stopped forming gas at 250–290 °C or kept it below 210 °C, emergency stops, and the short overlap with natural gas at the fuel switch-over.
- **Switch-on delay:** the four gas-line heaters come on 0–9 minutes after the air starts (typically 2–3 min), and a few hot-standby visits start up to 13 min late. Those waiting minutes are no longer counted as "off". This raises the heaters' running share by about 1 point (71 → 72 % in heat-up air, 97 → 98 % in hot standby) and changes no verdict. Most of the remaining off-time in heat-up air comes from two unusual heat-ups (13 Nov 2025 and 22 Mar 2026), when the heaters stayed off for hours.
- **How it was checked:** Page 3 lists which sensor stands for which device.

IF YOU REMEMBER ONE THING

The gas and air supplies match the page exactly. The small heaters switch on earlier, and the burner lights later, than page 6 shows.

Checked against the 10-second plant log (39 days), averaged to 1-minute steps and split by the project stage labels. Compared with page 6 of the plant walkthrough.

WHAT DIFFERS · WITH THE NUMBERS

Six corrections, from biggest to smallest

Each box shows what page 6 says, what the sensors show, and the change we suggest.

1 · GAS-LINE HEATERS · CLEARLY DIFFERENT

The small heaters are on from the very first start-up step

Page 6 shows the reformer heater, gas heater and off-gas heaters as “partly” on and the recycle heater EAOH 3 as off while the plant heats with air only. Their measured electric currents say they are running for about 7 of every 10 minutes in that step (72 %, with the 0–9 min switch-on delay at the start left out), at almost the same current as in hot standby. EAOH 3 draws the most of all, about 8 A, twice what it draws while making power.

MEDIAN CURRENT (A)	OFF	HEAT-UP AIR	HEAT-UP N_2+H_2	HOT STANDBY	FUEL SWITCH	PART LOAD	AIR-FLOW COOLING
Reformer heater ERH	0.0	3.2	3.2	3.2	5.2	5.1	0.0
Gas heater EAGH	0.0	3.0	3.2	2.9	4.8	4.8	0.0
Off-gas heater EAOH 1	0.0	3.3	3.4	3.3	4.6	3.0	0.0
Off-gas heater EAOH 2	0.0	3.1	3.2	3.1	4.5	2.9	0.0
Recycle heater EAOH 3	0.0	8.2	8.3	8.4	12.1	4.4	0.0

Suggested change: show all four heater groups as on from “Start-up: heat in air” through “Part load”. Show ERH as partly on (about two-thirds current) until the fuel switch-over. Replace the footnote “inferred, not measured” with “measured by current sensors IIR 3.1.1 / 5.1.1 / 5.1.2 / 5.1.3 / 6.1.1”.

Change made: the walkthrough now shows all four heater groups (ERH, EAGH, EAOH 1·2, EAOH 3) on from “Heat-up air” through “Part load”. Running below full power (partly) counts

as on; ERH is no longer shown as partly. The recycle heater EAOH 3 switches on 0–8 min after the air starts in 12 of 14 heat-up runs; the two runs without it (13 Nov 2025, 22 Mar 2026) were unusual starts.

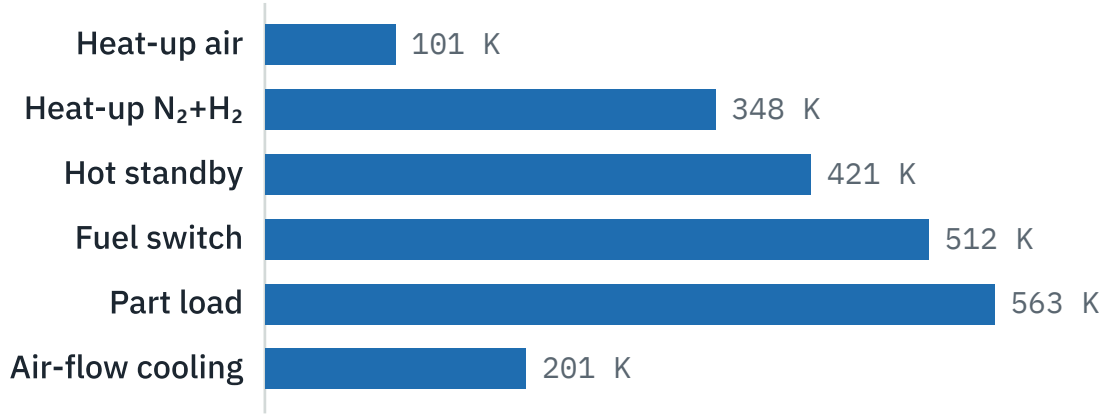
Switch-on delay: all four heater groups come on a few minutes after the air: typically 2–3 min, at most 9 min in heat-up air and 13 min in hot standby. Those minutes are now left out of the count (25–35 min per heater in heat-up, 15–25 min in standby). The running shares go up by about 1 point and no verdict changes.

2 · AIR PRE-HEATERS · ONE STEP APART IN 6 STAGES

The pre-heaters warm the air whenever air flows through a hot box

Page 6 shows the air pre-heaters as only partly working until the burner lights, and off during a trip. The air leaving them is 100 to 580 °C warmer than the air entering them in every stage that has air flow. They have no switch: any hot gas on the other side warms the air, including the warm air coming back from the stacks during start-up and air-flow cooling.

Air temperature rise across APH 1·2



Median of TIR 2.02 (outlet) minus TIR 2.1.1 / 2.1.2 (inlets), in the stages where air flows. With no air flowing (plant off, emergency stop) the pre-heaters heat nothing.

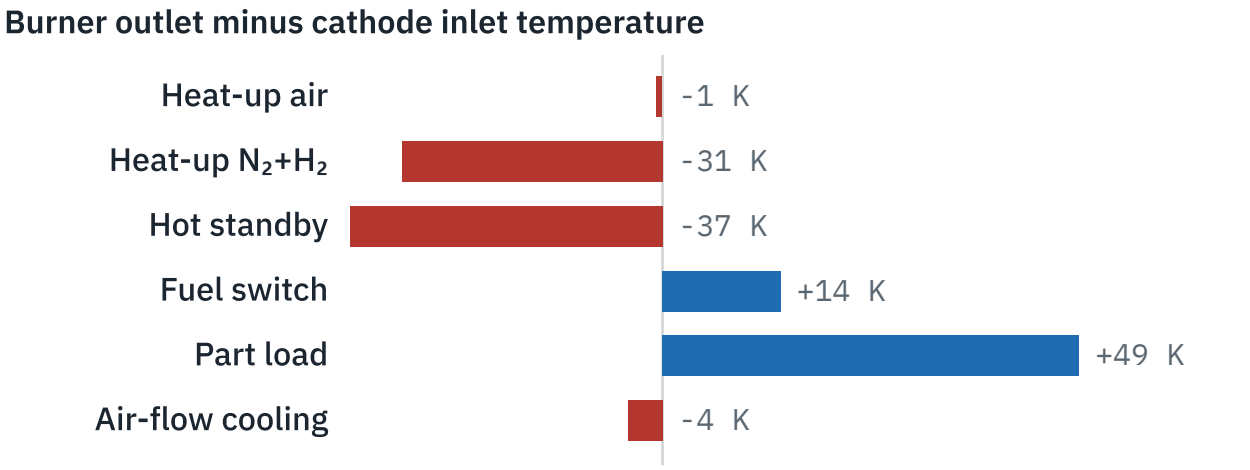
Suggested change: show APH as on in every stage that has air flow (all except “Plant off”), and change the label from “need hot exhaust” to “passive: heat whatever hot gas passes”.

3 · BURNER · ONE STEP APART IN 4 STAGES

No sign of burning until natural gas arrives

Page 6 shows the burner partly on while forming gas flows (start-up, hot standby, cooling). If the burner were burning, gas leaving it would be hotter than gas entering it. On forming

gas it is 31 to 37 °C cooler, so heat is being lost, not made. The burner gets hotter than its inlet only after the fuel switch-over (half of those minutes) and in part load (+49 °C).



Median of TIR 7.1.1 / 7.1.2 (burner outlet) minus TIR 5.4.1 / 5.4.2 (cathode gas into the burner). Red bars = burner colder than its inlet.

Suggested change: show the burner as off in forming-gas stages, air-flow cooling and plant off, partly on during the fuel switch-over, and on in part load. A small hydrogen flame could still be hidden by heat loss, so the label could read “no measurable burning”.

4 · COOLING COLUMNS · ONE STEP APART

Stacks, loads and ejector are already at zero when cooling starts

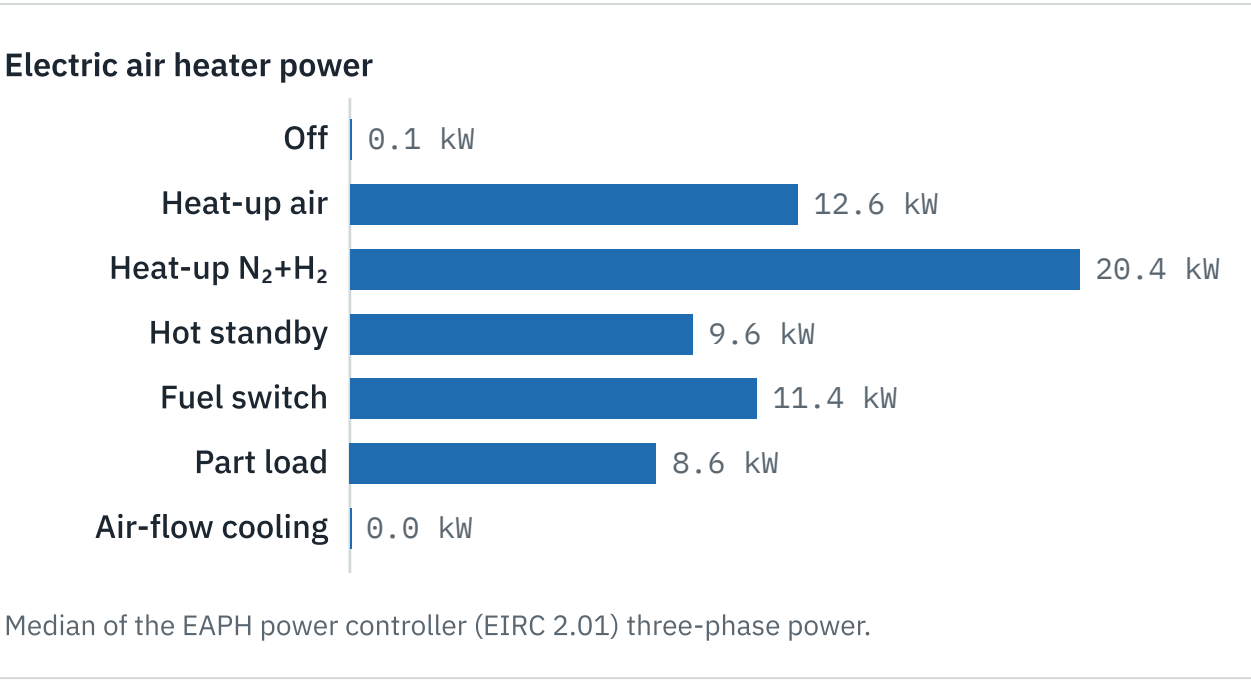
Page 6 shows the stacks, loads and ejector as partly on during cooling because the load ramps down. In the data, stack and load current are zero in every cooling minute, and the ejector drive pressure in 91 % of them. The ramp-down minutes are counted as part load by the stage labels, so cooling starts after the current has reached zero. Once the air is off (plant off) everything except the control system is off, as expected.

Suggested change: show stacks, loads and ejector as off in air-flow cooling and plant off, and move the sentence “load ramps to zero” to the end of “Part load”.

5 · SMALLER POINTS

Electric air heater and compressor

Electric air heater EAPH. Page 6 says it “works hard” in the air-only start-up. It runs nearly all the time there, but at a median of 12.6 kW; it reaches full effort (about 20 kW) only in the forming-gas part of the start-up. In standby and power it trims at about 9–11 kW. So the difference between “working hard” and “trimming” is about two to one, not ten to one.



Median of the EAPH power controller (EIRC 2.01) three-phase power.

Compressor. The 15 bar gas tank is filled during the last part of hot standby (a quarter of those minutes), before the fuel switch-over. Page 6 shows the compressor off until the switch-over.

Suggested change: label EAPH as “on, ~13 kW” in air-only start-up and “on, ~20 kW” in forming-gas start-up; show the compressor as partly on in hot standby.

6 · SAFETY TRIPS · FOLLOW THE SAFETY DESIGN

Two trip types, exactly as in the component sensor map

Page 6 had one trip column. The safety design (component sensor map, section 4) has two, and the 9 logged trips match them:

	EMERGENCY PURGE	EMERGENCY STOP
Caused by	Temperature, insulation, pressure or exhaust-fan trip	Gas detector, smoke or emergency-stop button
On	Air (2 × 3,000 Nl/min), N_2 (1,200 Nl/min), H_2 (60 Nl/min), control	Control only
Off	Natural gas and POx air (closed and vented), heaters, stacks, loads, compressor	Everything else, air and forming gas included
Logged after the trip	Air 6,000 · N_2 1,200 · H_2 60 · gas 0 · current 0	All flows 0 · current 0

Change made: the walkthrough now shows these two trip columns, as in the map.

IF YOU REMEMBER ONE THING

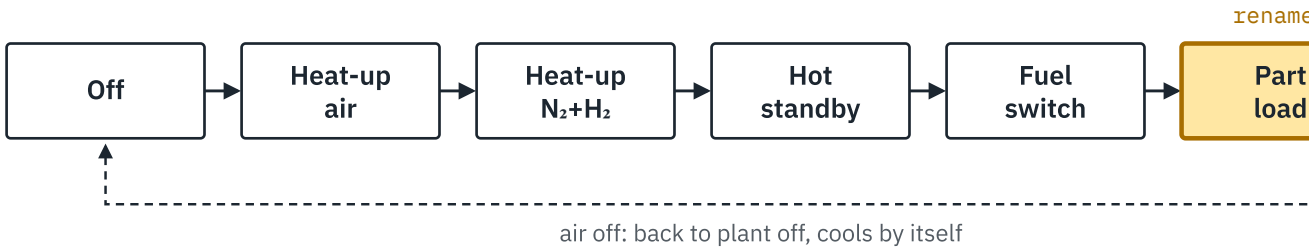
The heaters that keep the fuel lines warm start with the plant, not with the fuel; the burner only lights when natural gas arrives.

Checked against the 10-second plant log (39 days), averaged to 1-minute steps and split by the project stage labels. Compared with page 6 of the plant walkthrough.

STAGE CHANGES

Air-flow cooling, then plant off, and “Making power” is now Part load

Both changes fit the data. The plant cools in two clearly different ways, and power only ever follows a fuel switch-over, after a short wait.



The new stage order. Highlighted boxes are the renamed and new stages. A safety trip can interrupt any hot stage and is not drawn.

Two ways of cooling down

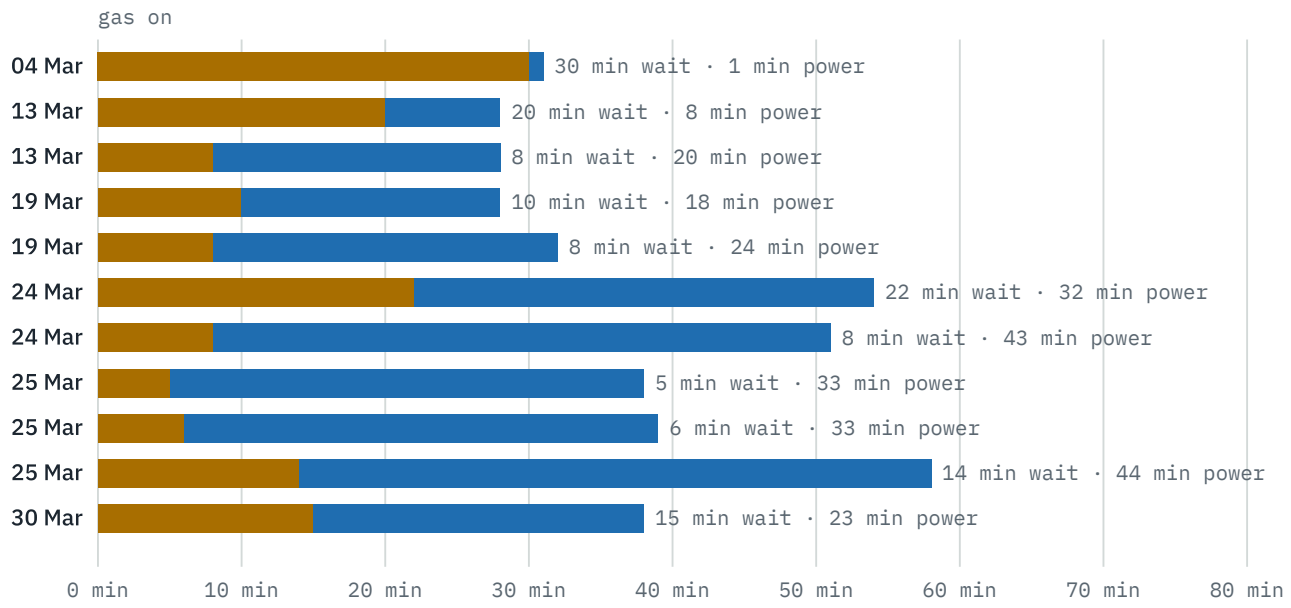
Inside the old shut-down stage the air supply is either running or switched off. With air it is **air-flow cooling**; once the air is off nothing runs, so it is simply **plant off**:

	AIR-FLOW COOLING	PLANT OFF (AIR OFF)
What happens	The air blowers keep pushing about 6,000 Nl/min of room air through the hot stacks	Air is off; the hot box loses heat to the room by itself
Minutes in the log	15,239	9,738 (about 3,200 of them still cooling, the rest cold)
Typical stack temperature	about 245 °C	about 65 °C while still cooling, then room temperature
Typical cooling speed	about 7 °C per hour (up to ~40 °C per hour with extra air)	under 1 °C per hour
Devices on	air supply, pre-heaters (passively), forming gas while still hot, control	control only

They differ in what runs and in how fast the stack cools. The gap in cooling speed (about 7 against under 1 °C per hour) matters when you plan how long the plant needs to reach a safe temperature. Part of that gap comes from the stack already being cooler during plant off, since a cooler stack loses heat more slowly anyway.

From fuel switch-over to part load

Power never starts on its own. All 279 minutes with stack current in the log had natural gas flowing, and the current always came **after** the gas was switched on. The fuel switch-over is the waiting time in between, while the reformer and stacks settle on the new fuel.



Each row is one natural-gas run that made power. Amber is the fuel switch-over (gas on, no current yet), blue is part load. The wait was 5–30 minutes, typically 10 minutes. Six more gas starts (1–18 minutes long) ended without any current.

That is why the fuel switch-over stays as its own stage before part load, and why it does not share the name: in those minutes the gas is on but no power is made.

IF YOU REMEMBER ONE THING

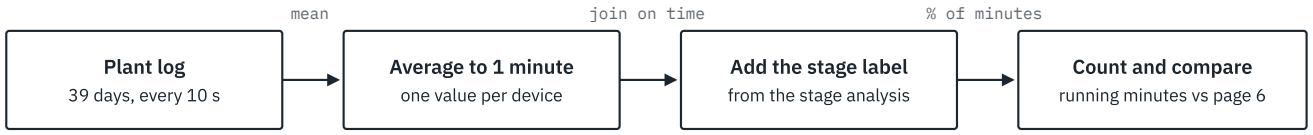
Gas on → about 10 minutes of waiting → part load. When the plant stops, it cools fast with air on and very slowly without it.

Checked against the 10-second plant log (39 days), averaged to 1-minute steps and split by the project stage labels. Compared with page 6 of the plant walkthrough.

HOW WE CHECKED

One sensor per device, one label per minute

For each device we picked the logged signal that shows it is really working, then counted in how many minutes of each stage that signal said “running”.



Rule for the data verdict: running in at least 60 % of a stage’s minutes = on; 10–60 %, or running at low power = partly; under 10 % = off.

Which signal stands for which device

DEVICE	SIGNAL USED AND THRESHOLD
Air supply	FIRC 1.1.1 + 1.1.2 flow
Air pre-heaters APH	air flowing and warmed TIR 2.1.x→2.02 by > 20 K (full: > 100 K)
Electric air heater EAPH	EIRC 2.01 3-phase power
Nitrogen line	FIRC 1.3 flow
Hydrogen line	FIRC 1.5 flow
Natural gas flow	FIRC 1.4 flow
Gas compressor	PT02 Haug pressure > 5 bar
POx air to reformer	FIRC 1.2 flow
Ejector (jet pump)	pumping: motive pressure > 0.5 bar; pass-through: ΔP > 1
Reformer heater ERH	IIR 5.1.3 current > 0.5 A
Gas heater EAGH	IIR 3.1.1 current > 0.5 A
Off-gas heaters EAOH 1·2	IIR 5.1.1/5.1.2 current > 0.5 A

DEVICE	SIGNAL USED AND THRESHOLD
Recycle heater EAOH 3	IIR 6.1.1 current > 0.5 A
Burner TOX	air flowing and TOX outlet hotter than cathode inlet by > 10 K
Stacks making power	string current > 1 A
Electronic loads	EIRC 4.1–4.5 current monitor > 1 A
Control · PLC · UPS	data logged at all

How page 6 stages map to the data labels

PAGE 6 STAGE	DATA LABEL USED	MINUTES
Plant off	S0 cold off + S1 warm off + S7 shut-down with the air off	9,738
Start-up: heat in air	S2 start-up while the gas phase is “air only”	3,376
Start-up: forming gas	S2 start-up while forming gas (N_2 + H_2) flows	4,891
Hot standby	S3 hot standby + S4 thermal adjustment, on forming gas	20,220
Fuel switch-over	natural gas flowing, no stack current, not shut-down or trip	207
Part load	S5 part load (rated power never reached)	265
Air-flow cooling	S7 shut-down while air flows (FIRC 1.1.1 + 1.1.2 above 20 Nl/min)	15,239
Safety trip: emergency purge	S8 trip windows after an emergency purge (flows only)	77
Safety trip: emergency stop	S8 trip windows after an emergency stop (flows only)	35

Limits

- **Forming gas in air-flow cooling:** judged against the stack temperature (T_stack): on in 92 % of the cooling minutes with the stack above 210 °C and in 6 % of those below, so the 210 °C rule explains why forming gas is on for only about half of all cooling minutes.
- **Switch-on delay of the gas-line heaters:** if ERH, EAGH, EAOH 1·2 or EAOH 3 is still off at the start of a heat-up, standby, fuel switch-over or part-load visit and comes on within 15 minutes, those minutes are a switch-on delay and are left out of the count instead of being counted as off. Longer waits are still counted as off.

- **Heater current columns:** the PLC copies (rCurrERH, rCurrEAGH, rCurrEAOH 1–3) are empty; the separate IIR current sensors carry the data and were used.
- **Safety trips** are shown as in the safety design (component sensor map, section 4). Only their air and gas flows were compared with the 9 logged trips.
- **Burner and pre-heaters** count as working only while air flows. With every valve shut, stored heat alone can make the outlet read warmer than the inlet.
- **Fuel switch-over and part load:** few minutes (207 and 265), because the plant made power on only a few days.
- **Plant off** now includes the minutes after shut-down when the air is off and the stack cools by itself; everything except control is off there too.
- **Air pre-heater outlet sensor** TIR 2.02 sits inside the warm box, so a little of its rise in quiet stages may be heat soaking from the box itself.

IF YOU REMEMBER ONE THING

Every verdict comes from a measured signal for that device, counted minute by minute in the same stage labels the rest of the project uses.

Checked against the 10-second plant log (39 days), averaged to 1-minute steps and split by the project stage labels. Compared with page 6 of the plant walkthrough.